

CLASSROOM CONTACT PROGRAMME

(ACADEMIC SESSION 2019-2020)

PRE-MEDICAL : ENTHUSIAST COURSE [PHASE-II]

TEST # 03

DATE : 23-06-2019

Test Type : Minor

Test Pattern : NEET

INSTRUCTIONS

Do not open this Test Booklet until you are asked to do so

1. Immediately fill in the form number on this page of the Test Booklet with Blue/Black Ball Point Pen. Use of pencil is strictly prohibited.
2. The candidates should not write their Form Number anywhere else (except in the specified space) on the Test Booklet/Answer Sheet.
3. The test is of **3 hours** duration.
4. The Test Booklet consists of **180** questions. The maximum marks are **720**.
5. There are **three** parts in the question paper A,B,C consisting of **Physics, Chemistry** and **Biology** having **45 questions** each in Physics & Chemistry and **90 questions** in Biology part of equal weightage. Each question is allotted 4 (four) marks for **correct** response.
6. **One Fourth** mark will be deducted for indicated incorrect response of each question. **No deduction** from the total score will be made if no response is indicated for an item in the Answer Sheet.
7. Use **Blue/Black Ball Point Pen only** for writing particulars/markings responses on **Side-1** the Answer Sheet. **Use of pencil is strictly prohibited.**
8. No candidate is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone any electronic device etc, except the Identity Card inside the examination hall/room.
9. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
10. On completion of the test, the candidate must hand over the Answer Sheet to the invigilator on duty in the Room/Hall. **However, the candidate are allowed to take away this Test Booklet with them.**
11. **Do not fold or make any stray marks on the Answer Sheet.**

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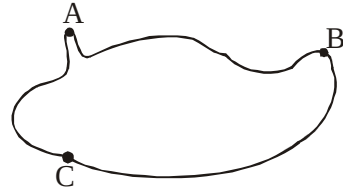
SPACE FOR ROUGH WORK

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BEWARE OF NEGATIVE MARKING

1. Figure shows a charged conductor of irregular shape. If σ_A , σ_B and σ_C are the surface charge densities at A, B and C respectively, then :-

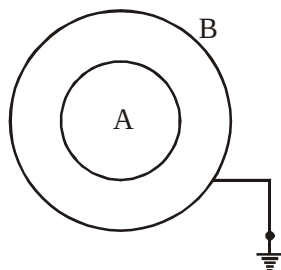
- (1) $\sigma_A = \sigma_B = \sigma_C$
(2) $\sigma_A < \sigma_B < \sigma_C$
(3) $\sigma_A > \sigma_B > \sigma_C$
(4) $\sigma_A = \sigma_B < \sigma_C$



2. A spherical conducting shell has a charge q . If a point charge Q is placed at the centre, then the charge on inner and outer surface of the shell respectively are :-

- (1) 0, Q (2) $-Q$, $Q + q$
(3) q , $Q - q$ (4) Q , $Q + q$

3. A is a spherical conductor placed concentrically inside a earthed hollow spherical conductor B. If the conductor A is given a charge of $+Q$ then the electric field is non-zero :-



- (1) inside A
(2) outside B
(3) in between A and B
(4) none of these

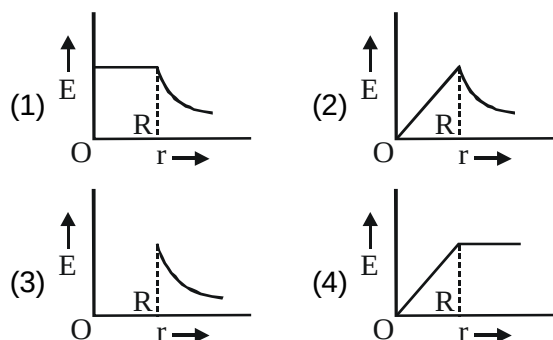
4. An electric dipole is placed in a uniform field, the net force and net torque on the dipole is :-

- (1) force = 0, torque may not be zero
(2) force may not be zero, torque = 0
(3) force and torque both may not be zero
(4) force = 0, torque = 0

5. An uncharged aluminum block has a cavity within it. The block is placed in a region where a uniform electric field is directed upward. Which of the following is a correct statement describing conditions in the interior of the block's cavity?

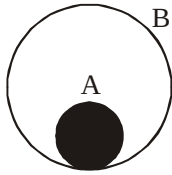
- (1) The electric field in the cavity is directed upward
(2) The electric field in the cavity is directed downward.
(3) There is no electric field in the cavity
(4) The electric field in the cavity is of varying magnitude and is zero at the exact center.

6. The electric field due to a uniformly charged sphere of radius R as a function of the distance from its centre is represented graphically by :-

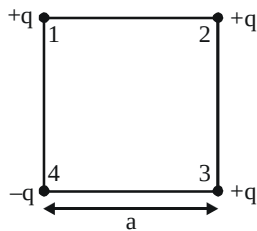


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7. A charged conductor A is kept inside a hollow conductor B in contact with B as shown in figure. Then :-

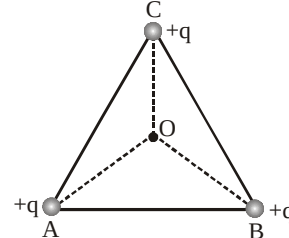


- (1) opposite charge is induced on the outer surface of B
 - (2) same kind of charge is induced on the outer surface of B
 - (3) the whole charge of A gets transferred to the outer surface of B
 - (4) the whole charge of A gets transferred to the inner surface of B.
8. The work done required to put the four charges together at the corners of a square of side a , as shown in the figure is :-



- (1) $\frac{1}{4\pi\epsilon_0} \frac{q^2}{a}$
 - (2) $-\frac{2.6}{4\pi\epsilon_0} \frac{q^2}{a}$
 - (3) $+\frac{2.6}{4\pi\epsilon_0} \frac{q^2}{a}$
 - (4) zero
9. At any point on the perpendicular bisector of the line joining the two equal and opposite charges :-
- (1) the electric field is zero
 - (2) the electric potential is zero
 - (3) the electric field is perpendicular to the line joining the two charges
 - (4) the electric potential decreases with the distance from the centre of the charges.

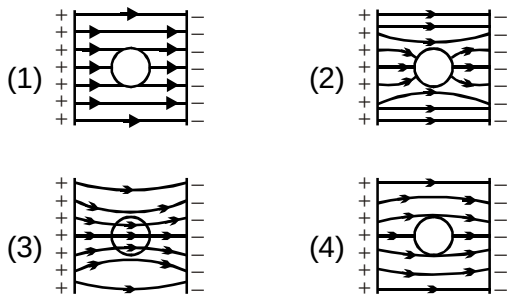
10. Like charges of equal magnitude q are kept at the vertices of an equilateral triangle ABC as shown in the fig. The electric intensity E and potential V , at the centroid O , are respectively :-



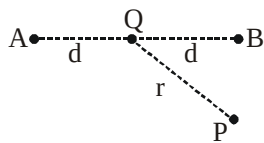
- (1) $E = 0, V = 0$
 - (2) $E = 0, V \neq 0$
 - (3) $E \neq 0, V = 0$
 - (4) $E \neq 0, V \neq 0$
11. A hollow metallic sphere of radius 8 cm is given a charge of 3.2×10^{-9} C. The electric potential at a distance 5 cm from its centre is :-
- (1) 3.6×10^2 V
 - (2) 9×10^2 V
 - (3) zero
 - (4) 28.8×10^{-9} V
12. If E and V represent electric field and electric potential at a point in space respectively, then :-
- (1) if $E = 0$, V may be zero or constant
 - (2) if $E = 0$, V must be zero
 - (3) if $V = 0$, E must be zero
 - (4) if $E = \text{constant}$, V must be zero
13. If an electron is thrown in the direction of electric field then :-
- (1) its kinetic energy decreases but potential energy will increase
 - (2) both kinetic and potential energy will increase
 - (3) both kinetic and potential energy will decrease
 - (4) kinetic energy will increase but potential energy will decrease

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14. A sample of HCl gas is placed in an electric field of $2.5 \times 10^4 \text{ NC}^{-1}$. The dipole moment of each HCl molecule is $3.4 \times 10^{-30} \text{ C m}$. The maximum torque acting on each molecule :-
 (1) $2.5 \times 10^{-30} \text{ N m}$ (2) $8.5 \times 10^{-30} \text{ N m}$
 (3) $8.5 \times 10^{-26} \text{ N m}$ (4) $8.5 \times 10^4 \text{ N m}$
15. An uncharged metal sphere is placed between two equal and oppositely charged metal plates. The nature of lines of force will be :-



16. The work done by external agent in taking a unit positive charge from P to A is W_A and from P to B is W_B :-

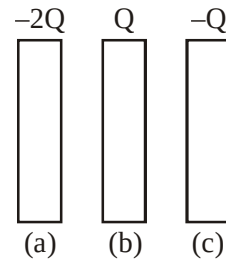


Then

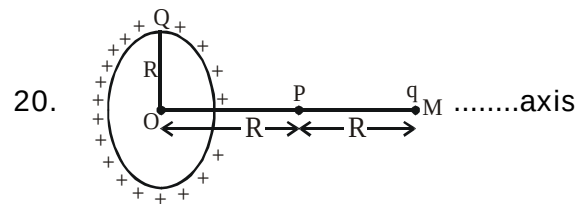
- (1) $W_A > W_B$ (2) $W_A < W_B$
 (3) $W_A = W_B$ (4) $W_A + W_B = 0$
17. Two isolated charged conducting spheres of radii a and b produces the same electric field on their respective surfaces, the ratio of electric potentials on their surfaces is :-

- (1) $\frac{a}{b}$ (2) $\frac{b}{a}$ (3) $\frac{a^2}{b^2}$ (4) $\frac{b^2}{a^2}$

18. For a charged conductor which one statement is wrong :-
 (1) Electric line of force is perpendicular to surface
 (2) Electric field inside the surface is zero
 (3) Electric field is uniform on its surface
 (4) No charge exist on outer surface of conductor
19. Three identical conducting plates are shown in figure. Findout the charge on the right face of the plate b.



- (1) $-Q$ (2) $\frac{Q}{2}$
 (3) Q (4) Zero



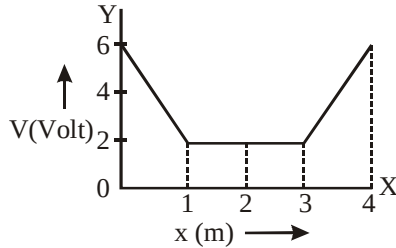
20.

A charge Q is uniformly distributed on a ring of radius R another charge q is placed at point M on the axis of ring if net potential at P point is zero then :-

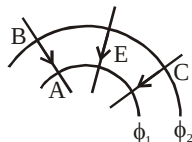
- (1) $q = \frac{Q}{\sqrt{2}}$ (2) $q = \frac{-Q}{\sqrt{2}}$
 (3) $q = \frac{-Q}{\sqrt{3}}$ (4) $q = \frac{-Q}{2}$

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21. The variation of electric potential with distance from a fixed point is shown in figure. What is the value of electric field at $x = 0.5$ m (in N/C) is :



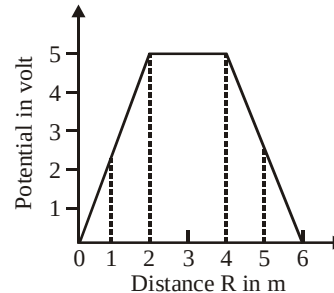
- (1) Zero (2) 3 (3) 4 (4) 2
22. A given charge situated at a distance r from an electric dipole on its axis experiences a force F . If the distance of the charge from the dipole is doubled, the force acting on the charge will be:-
- (1) $4F$ (2) $F/2$
(3) $F/4$ (4) $F/8$
23. In moving from A to B along an electric field line, the electric field does 6.4×10^{-19} J of work on an electron. If ϕ_1 , ϕ_2 are equipotential surfaces, then the potential difference ($V_C - V_A$) is :-



- (1) $-4V$ (2) $4V$
(3) zero (4) $64V$
24. Which of the following is sufficient condition for finding the electric flux ϕ_E through a closed surface?

- (1) If the magnitude of $\vec{E}$ is known everywhere on the surface
(2) If the total charge inside the surface is specified
(3) If the total charge outside the surface is specified
(4) Only if the location of each point charge inside the surface is specified

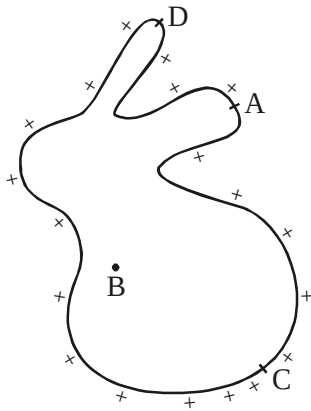
25. The potential of the electric field produced by a point charge at any point (x, y, z) is given by; $V = 3x^2 + 5$, where x, y, z are in metres and V is in volts. The intensity of the electric field at $(2, 1, 0)$ is :-
- (1) $+17 \text{ Vm}^{-1}$ (2) -17 Vm^{-1}
(3) $+12 \text{ Vm}^{-1}$ (4) -12 Vm^{-1}
26. What is the angle between electric field and equipotential surface ?
- (1) 90° always (2) 0° always
(3) 0° to 90° (4) 0° to 180°
27. The variation of potential with distance R from a fixed point is as shown in the figure. The electric field at $R = 5$ m is :



- (1) 2.5 V/m (2) -2.5 V/m
(3) $(2/5) \text{ V/m}$ (4) $-(2/5) \text{ V/m}$
28. There are 10 units of charge at the centre of a circle of radius 1.0 m. The work done in moving 1 unit of charge once around the circle is :-
- (1) zero (2) 10 units
(3) 100 units (4) 150 units

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29. For the isolated charged conductor of figure, the electric fields at points A, B, C and D are E_A , E_B , E_C and E_D respectively. Then :-

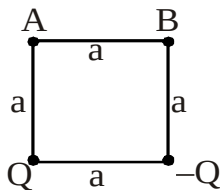


- (1) $E_A = E_B > E_C > E_D$
 (2) $E_D > E_C > E_B = E_A$
 (3) $E_D > E_A > E_C$ but $E_B = 0$
 (4) $E_B = 0$, $E_A = E_C = E_D$
30. An electric dipole is placed at an angle of 30° with an electric field of intensity 2×10^5 N/C. It experiences a torque equal to 4 Nm. Calculate the charge on the dipole if the dipole length is 2 cm.
 (1) 8 mC (2) 4 mC (3) $8 \mu\text{C}$ (4) 2 mC
31. Work done to move a charge q_0 from A to B will be :-
 (1) Zero

(2) $\frac{2KQq_0}{a} \left[-1 + \frac{1}{\sqrt{2}} \right]$

(3) $\frac{KQq_0}{a} \left[1 + \frac{1}{\sqrt{2}} \right]$

(4) $\frac{KQq_0}{a} \left[1 - \frac{1}{\sqrt{2}} \right]$



32. A point Q lies on the perpendicular bisector of an electric dipole of dipole moment p. If the distance of Q from the dipole is r (much larger than the size of the dipole) then electric field at Q is proportional to :-

- (1) p^{-1} and r^{-2} (2) p and r^{-2}
 (3) p^2 and r^{-3} (4) p and r^{-3}

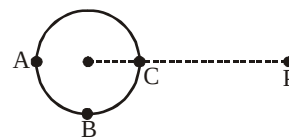
33. If the electric field is given by $(5\hat{i} + 4\hat{j} + 9\hat{k})$, the electric flux through a surface of area 20 unit lying in the Y-Z plane will be :-

- (1) 100 unit (2) 80 unit
 (3) 180 unit (4) 20 unit

34. Two conductors having the same type of charges are connected by a conducting wire. There would not be any change in amount of charges on them if :-

- (1) they have the same potential
 (2) they have the same amount of charge
 (3) they have the same capacity
 (4) they have the same shape

35. A hollow conducting sphere is placed in an electric field produced by a point charge placed at P as shown in figure. Let V_A , V_B , V_C be the potentials at points A, B and C respectively. Then :-



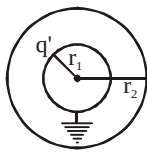
- (1) $V_C > V_B$ (2) $V_B > V_C$
 (3) $V_A > V_B$ (4) $V_A = V_C$

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36. If a unit positive charge is taken from one point to another over an equipotential surface, then :-

- (1) Work is done on the charge
- (2) Work is done by the charge
- (3) Work done is constant but not zero
- (4) No work is done

37. The concentric spheres are of radii r_1 and r_2 . The outer sphere is given a charge q . The charge q' on the inner sphere will be (inner sphere is grounded):-



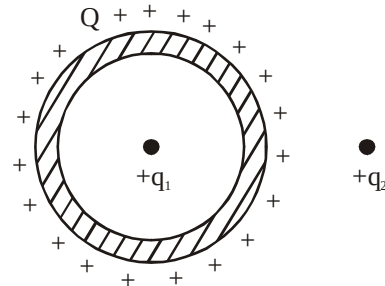
- (1) q
- (2) $-q$
- (3) $-q \frac{r_1}{r_2}$
- (4) zero

38. Positive and negative point charges of equal magnitude are kept at $\left(0, 0, \frac{a}{2}\right)$ and

$\left(0, 0, -\frac{a}{2}\right)$ respectively. The work done by the electric field when another positive point charge is moved from $(-a, 0, 0)$ to $(0, a, 0)$ is :-

- (1) positive
- (2) negative
- (3) zero
- (4) dependent on the path connecting the initial and final positions

39. A thin metallic spherical shell contains a charge Q on its surface. A point charge q_1 is placed at the center of the shell, and another charge q_2 is placed outside the shell. All the three charges are positive. Then the force on charge q_1 is:-



- (1) toward right
- (2) toward left
- (3) zero
- (4) none of these

40. An electron (of charge $-e$) revolves around a long wire with uniform charge density λ in a circular path of radius r . Its kinetic energy is given by :-

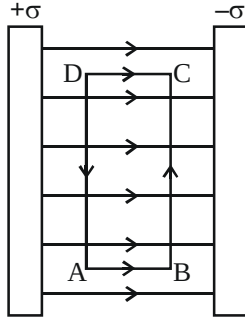
- (1) $\frac{\lambda e}{2\pi\epsilon_0 r}$
- (2) $\frac{\lambda e}{4\pi\epsilon_0 r}$
- (3) $\frac{\lambda e}{2\pi\epsilon_0}$
- (4) $\frac{\lambda e}{4\pi\epsilon_0}$

41. Consider a thin spherical shell of radius R consisting of uniform surface charge density σ . The electric field at a point of distance x from its centre and outside the shell is :-

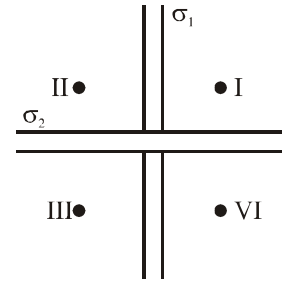
- (1) Inversely proportional to σ
- (2) Directly proportional to x^2
- (3) Directly proportional to R
- (4) Inversely proportional to x^2

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42. Consider identical but oppositely charged two large parallel plates, with charge density σ . A small charge q_0 is moved along the rectangular path ABCDA where side AB = x and side BC = y. Then the incorrect statement is :-



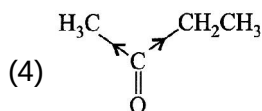
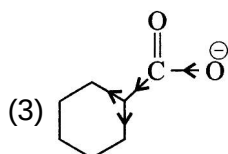
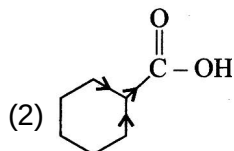
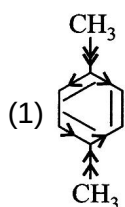
- (1) work done by electric field along path AB is positive and equal to $q_0\sigma x/\epsilon_0$
 (2) work done by electric field along path BC is non zero
 (3) work done by electric field along path ABCDA is zero
 (4) None of the above
43. The electric potential at a point due to an electric dipole is :-
 (1) $k(\vec{p} \cdot \vec{r}/r^3)$ (2) $k(\vec{p} \cdot \vec{r}/r^2)$
 (3) $k(\vec{p} \times \vec{r}/r^3)$ (4) $k(\vec{p} \times \vec{r}/r^2)$
44. Two infinite nonconducting thin sheets having charge densities σ_1 and σ_2 are placed in two perpendicular planes whose two-dimensional view is shown in Fig. The charges are distributed uniformly on the sheets in electrostatic equilibrium condition. Four points are marked I, II, III and IV. The electric field intensities at these points are $\vec{E}_1, \vec{E}_2, \vec{E}_3$ and $\vec{E}_4$, respectively. The correct expression for the electric field intensities is:-



- (1) $|\vec{E}_1| = |\vec{E}_2| = \frac{\sqrt{\sigma_1^2 + \sigma_2^2}}{2\epsilon_0} \neq |\vec{E}_4|$
- (2) $|\vec{E}_2| \neq |\vec{E}_4| = \frac{\sqrt{\sigma_1^2 + \sigma_2^2}}{2\epsilon_0}$
- (3) $|\vec{E}_1| = |\vec{E}_2| = |\vec{E}_3| = |\vec{E}_4| = \frac{\sqrt{\sigma_1^2 + \sigma_2^2}}{2\epsilon_0}$
- (4) none of these
45. Two opposite and equal charge 4×10^{-8} coulomb when placed 2×10^{-2} cm away, form a dipole. If this dipole is placed in an external electric field 4×10^8 newton/coulomb, the value of maximum torque and the work done in rotating it through 180° will be :-
 (1) 64×10^{-4} Nm and 64×10^{-4} J
 (2) 32×10^{-4} Nm and 32×10^{-4} J
 (3) 64×10^{-4} Nm and 32×10^{-4} J
 (4) 32×10^{-4} Nm and 64×10^{-4} J

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46. Inductive effect involves:
 (1) Delocalisation of σ -electrons
 (2) Partial displacement of σ -electrons
 (3) Delocalisation of π -electrons
 (4) Displacement of lone pair electrons.
47. What is the correct increasing order of extent of deprotonation of the following acids in its aqueous solution?
 (I) $(\text{CH}_3)_2\text{CHCOOH}$
 (II) $\text{CH}_3\text{CH}_2\text{COOH}$
 (III) $\text{Cl}_2\text{CHCH}_2\text{COOH}$
 (IV) $\text{CH}_3\text{CCl}_2\text{COOH}$
 (1) $\text{II} < \text{I} < \text{III} < \text{IV}$ (2) $\text{IV} < \text{III} < \text{I} < \text{II}$
 (3) $\text{I} < \text{II} < \text{III} < \text{IV}$ (4) $\text{I} < \text{IV} < \text{III} < \text{II}$
48. Decreasing -I effect of given groups is:
 (i) CN (ii) NO_2
 (iii) NH_2 (iv) F
 (1) $\text{iii} > \text{ii} > \text{i} > \text{iv}$ (2) $\text{ii} > \text{iii} > \text{iv} > \text{i}$
 (3) $\text{iii} > \text{ii} > \text{iv} > \text{i}$ (4) $\text{ii} > \text{i} > \text{iv} > \text{iii}$
49. Which of the following alkyl groups has the maximum +I effect?
 (1) $(\text{CH}_3)_2\text{CH}-$ (2) $(\text{CH}_3)_3\text{C}-$
 (3) CH_3CH_2- (4) CH_3-
50. In which of the following species, incorrect direction of Inductive effect is shown?



51. Which of the following acids has the smallest dissociation constant?
 (1) $\text{CH}_3\text{CHFCH}_2\text{COOH}$
 (2) $\text{FCH}_2\text{CH}_2\text{COOH}$
 (3) $\text{BrCH}_2\text{CH}_2\text{COOH}$
 (4) $\text{CH}_3\text{CHBrCH}_2\text{COOH}$
52. Assign number 1 for least to 4 for most to indicate the relative base strength of the following

I	II	III	IV
$\text{C}_6\text{H}_5\text{NH}_2$	$\text{p-NO}_2\text{C}_6\text{H}_4\text{NH}_2$	$\text{m-NO}_2\text{C}_6\text{H}_4\text{NH}_2$	$\text{p-CH}_3\text{OC}_6\text{H}_4\text{NH}_2$
I	II	III	IV
(1) 3	1	2	4
(2) 1	2	3	4
(3) 2	3	4	1
(4) 4	1	3	2

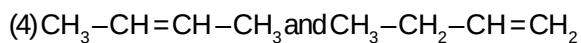
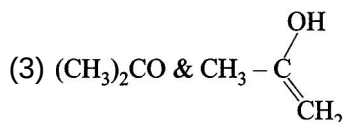
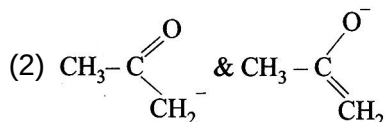
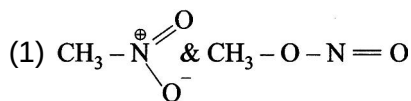
53. , pK_a value of the compound

decreases if X is:

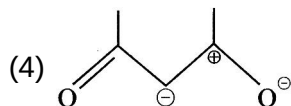
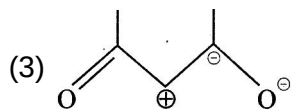
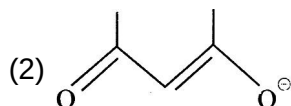
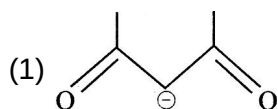
- (1) $-\text{NO}_2$
 (2) $-\text{NH}_2$
 (3) $-\text{OH}$
 (4) $-\text{OCH}_3$

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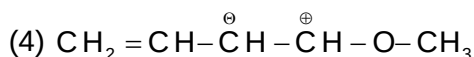
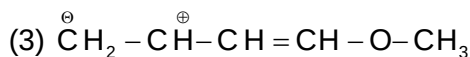
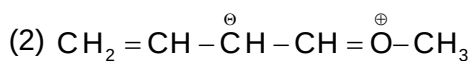
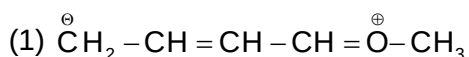
54. Which of the following pairs are resonating structures?



55. Which is the most stable resonating structure?



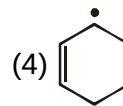
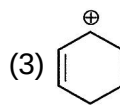
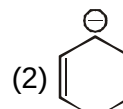
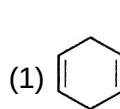
56. Which of the following resonating structures of 1-methoxy-1, 3-butadiene is least stable?



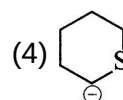
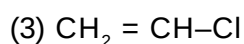
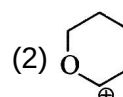
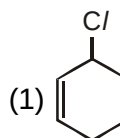
57. Resonance energy is:

- (1) The potential energy difference between most stable resonating structure and resonance hybrid
- (2) The potential energy difference between the least stable resonating structure and resonance hybrid
- (3) The potential energy difference between the least stable and most stable resonating structure
- (4) None

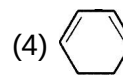
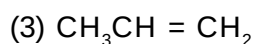
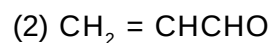
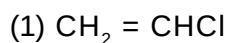
58. Which of the following species cannot show resonance?



59. Resonance is not possible in:

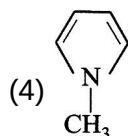
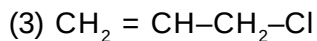
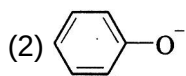
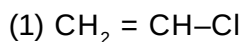


60. Which does not have conjugate system?



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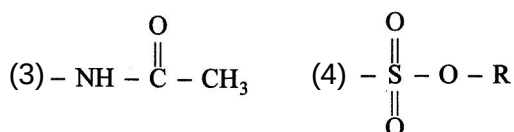
61. The compound which is not resonance stabilised



62. Which of the following group show +m effect?



63. Which of the following group show -m effect?



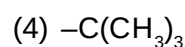
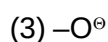
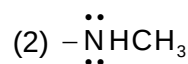
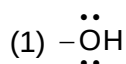
64. Which of the following group show +m > -I effect?



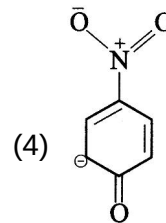
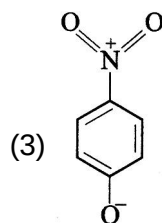
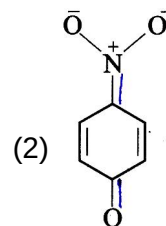
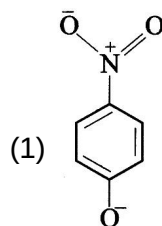
65. Which of the following group show -m and -I effect?



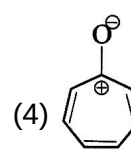
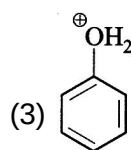
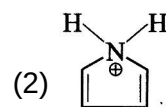
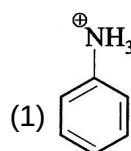
66. +m and +I both effects are shown by :



67. The most unlikely representation of resonance structures of p-nitrophenoxide ion is:

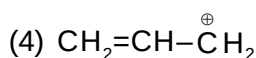
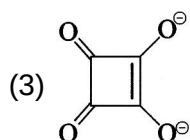
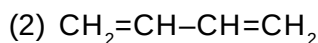
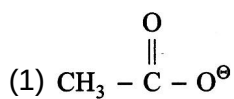


68. In which delocalisation of positive charge is possible?

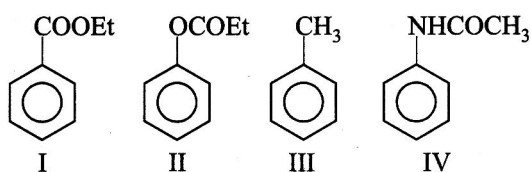


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69. Which of the following species does not have equally contributing resonating structures?

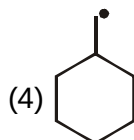
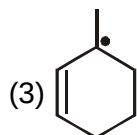
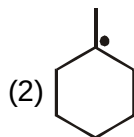
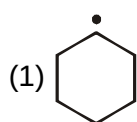


70. The correct order of electron density in aromatic ring of following compounds is:

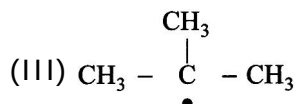
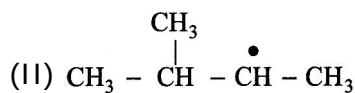
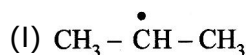


- (1) $\text{IV} > \text{III} > \text{II} > \text{I}$ (2) $\text{I} > \text{II} > \text{III} > \text{IV}$
(3) $\text{IV} > \text{II} > \text{I} > \text{III}$ (4) $\text{IV} > \text{II} > \text{III} > \text{I}$

71. Most stable radical among the following is:



72. Arrange the following in increasing order of number of hyperconjugation structures.

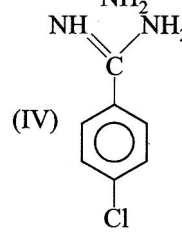
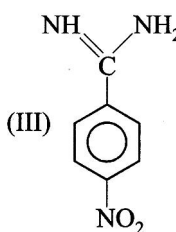
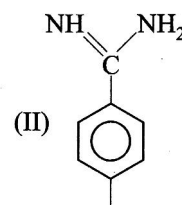
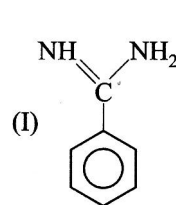


- (1) $\text{I} < \text{II} < \text{III}$ (2) $\text{II} < \text{I} < \text{III}$
(3) $\text{III} < \text{II} < \text{I}$ (4) $\text{II} < \text{III} < \text{I}$

73. Arrangement of $(\text{CH}_3)_3\text{C}^-$, $(\text{CH}_3)_2\text{CH}^-$, CH_3CH_2^- when attached to benzyl or an unsaturated group in increasing order of inductive effect is :

- (1) $(\text{CH}_3)_3\text{C}^- < (\text{CH}_3)_2\text{CH}^- < \text{CH}_3\text{CH}_2^-$
(2) $\text{CH}_3\text{CH}_2^- < (\text{CH}_3)_2\text{CH}^- < (\text{CH}_3)_3\text{C}^-$
(3) $(\text{CH}_3)_2\text{CH}^- < (\text{CH}_3)_3\text{C}^- < \text{CH}_3\text{CH}_2^-$
(4) $(\text{CH}_3)_3\text{C}^- < \text{CH}_3\text{CH}_2^- < (\text{CH}_3)_2\text{CH}^-$

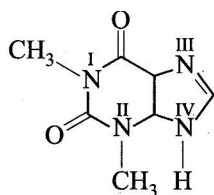
74. Select the decreasing order of relative basic strengths of following species:



- (1) $\text{II} > \text{IV} > \text{I} > \text{III}$ (2) $\text{III} > \text{I} > \text{IV} > \text{II}$
(3) $\text{III} > \text{IV} > \text{I} > \text{II}$ (4) $\text{II} > \text{I} > \text{IV} > \text{III}$

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75. The correct increasing order of basic strength of the labeled N is:

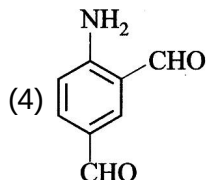
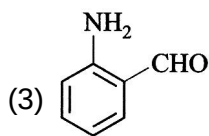
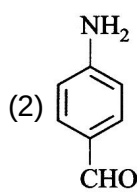
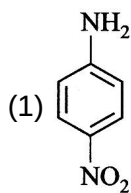


- (1) I < II < III < IV
(2) I < II < IV < III
(3) I < III < IV < II
(4) IV < III < II < I

76. Among the following compounds, the strongest acid is :

- (1) $\text{HC} \equiv \text{CH}$ (2) C_6H_6
(3) C_2H_6 (4) CH_3OH

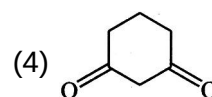
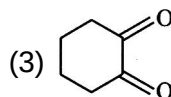
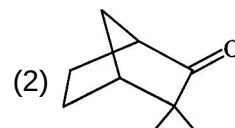
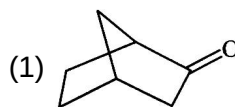
77. Which is most basic in the following compounds?



78. An isomer of ethanol is:

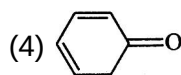
- (1) methanol
(2) diethyl ether
(3) acetone
(4) dimethyl ether

79. Among the following compounds, one that will not show keto-enol tautomerism is

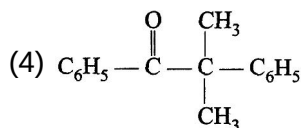
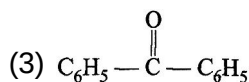
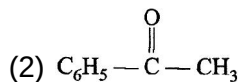
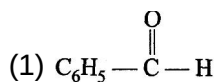


80. Maximum enolisation takes place in

- (1) CH_3COCH_3
(2) $\text{CH}_2\text{COCH}_2\text{CHO}$
(3) $\text{CH}_3\text{COCH}_2\text{COCH}_3$



81. Keto-enol tautomerism is observed in

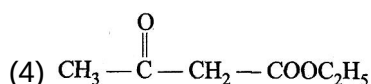
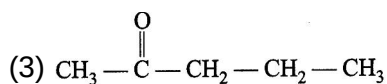
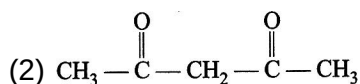
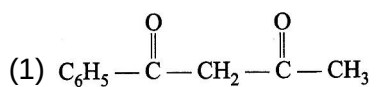


82. Tautomerism will be exhibited by

- (1) $(\text{CH}_3)_2\text{NH}$
(2) $(\text{CH}_3)_3\text{CNO}$
(3) R_3CNO_2
(4) RCH_2NO_2

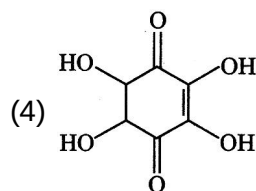
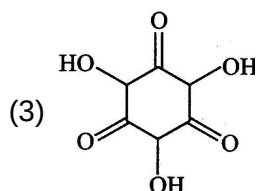
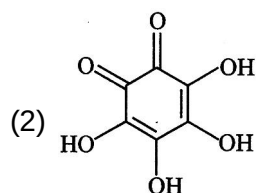
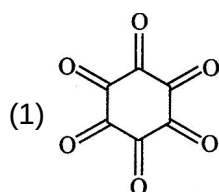
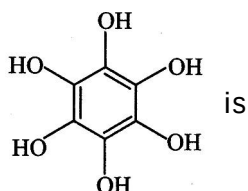
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83. Which among the following compounds will give maximum enol content in solution?



84. The most stable keto isomer of the

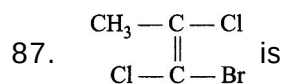
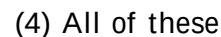
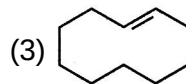
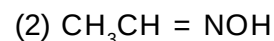
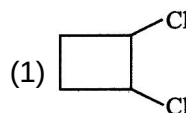
following compound



85. The number of isomers for the compound with molecular formula C_2BrClFI is

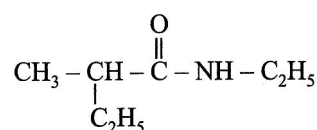
- (1) 3 (2) 4 (3) 5 (4) 6

86. Which will show geometrical isomerism?



- (1) trans (2) Z
(3) both correct (4) none of these

88. The IUPAC name of the compound is:

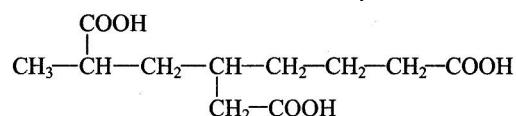


- (1) N-Ethyl-2-ethylpropanamide
(2) N-Ethyl-2-methylbutanamide
(3) N-Ethyl-2-methylpropanecarboxamide
(4) N-Ethyl-2-ethylethanecarboxamide

89. IUPAC name of $\text{H}_3\text{C}-\underset{\text{C}_2\text{H}_5}{\text{CH}}-\text{CH}_2-\underset{\text{OH}}{\text{CH}}-\text{CH}_2\text{Cl}$

- (1) 1-chloro-4-methylhexan-2-ol
(2) 1-chloro-4-methylhexan-2-al
(3) 1-chloro-4-ethylpentan-2-ol
(4) 1-chloro-2-hydroxy-4-methylhexane

90. IUPAC name of the compound



- (1) 2-methyl-4-ethanoic octane-1, 8-octanedioic acid
(2) 2-methyl-4-(carboxy methyl) octane-1, 8-dioic acid
(3) 2-methyl-1, 1', 6-heptane tricarboxylic acid
(4) None of these

SPACE FOR ROUGH WORK

91. What % of trihybrid will be obtained by the cross of trihybrid plant with a recessive plant :-
 (1) 25% (2) 12.5%
 (3) $\frac{1}{64} \times 100$ (4) $\frac{3}{8} \times 100$
92. The genes for male sterility & albinism in plants, are found respectively in :-
 (1) Mitochondria & Plastid
 (2) Plastid & Mitochondria
 (3) Plastid & Plastid
 (4) None of the above
93. Skin colour in human, is controlled by three gene pairs, then what is the number of intermediate & parental combinations in F_2 generation respectively :-
 (1) 2, 62 (2) 62, 2
 (3) 2, 14 (4) 14, 2
94. Which of the following type of inheritance can we find out with the help of reciprocal cross?
 (1) Cytoplasmic inheritance
 (2) Gene inheritance
 (3) Maternal inheritance
 (4) (1) & (3) both
95. Genes a, b and c assort independently and are recessive to their respective alleles A, B and C. Two triple heterozygous AaBbCc individual are crossed what is the probability that a given offspring will be genotypically homozygous for all three dominant alleles?
 (1) 2/64 (2) 27/64
 (3) 1/64 (4) 6/64
96. What is true for law of independent assortment?
 (1) Applicable to all the dominant alleles
 (2) Applicable to all genes on the same chromosome
 (3) four type of gametes are formed in dihybrid
 (4) Applicable to all recessive alleles
97. A dihybrid plant is with genotype GgTt. In this plant first gene pair is codominant on selfing of this plant what is phenotypic ratio?
 (1) 2 : 1 (2) 3 : 1
 (3) 3 : 6 : 3 : 1 : 2 : 1 (4) 6 : 3 : 2 : 1
98. Which will not the genotype of pollen grain of a plant with genotype HHbbDdnnMMaa?
 (1) HbdNMa (2) HbDnMa
 (3) HbdnMa (4) All the above
99. Choose the correct statement from the given below :-
 (1) % of heterozygosity increases due to selfing
 (2) Bagging technique was used by Mendel to prevent unwanted cross pollination.
 (3) When cross between red flowered & white flowered *Mirabilis jalapa* takes place then only red flowered plants are obtained in F_1 due to dominance.
 (4) Mendel used sweet pea plant for his experiments
100. The allele, which is unable to express its effect in the presence of another is called :-
 (1) Co-dominant
 (2) Supplementary
 (3) Complementary
 (4) Recessive
101. Mendel observed that all the F_1 progeny plants
 (1) Resembled either one of the parents
 (2) Resembled neither of the parents
 (3) Resembled both of the parents
 (4) Shows 3 : 1 ratio
102. The segregation of alleles is a random process and so there is a _____ chance of a gamete containing either allele :-
 (1) 25% (2) 50%
 (3) 75% (4) 100%

103. In the inheritance of flower colour in dog flower plant, the F_1 had a phenotype that
 (1) Resembles both of the parents
 (2) Did not resembles either of the two parents
 (3) Resemble with only one parent
 (4) (1) & (3) both
104. The three different alleles of human ABO blood types will produce how many genotypes & phenotypes respectively :-
 (1) 4 & 6 (2) 6 & 4
 (3) 6 & 6 (4) 4 & 4
105. "When two pairs of traits are combined in a hybrid, segregation of one pair of characters is independent of the other pair of characters".
 This explains :
 (1) Law of dominance
 (2) Law of segregation
 (3) Law of independent assortment
 (4) Postulate to paired factors
106. Other than pea plants it was found that sometimes the F_1 had a phenotype that did not resemble either of the two parents and was in between the two. Due to :-
 (1) Complete dominance
 (2) Incomplete dominance
 (3) Co-dominance
 (4) Complementary gene interaction
107. Which statement is true?
 (1) Characters segregate while formation of gamete
 (2) All characters show true dominance
 (3) The characters are always blend in heterozygous condition
 (4) Mendel's also worked on rice plant
108. Which of the following condition is true for co-dominance :-
 (1) Phenotype of F_1 resembled either of the two parents
 (2) Phenotype of F_1 did not resemble either of two parents
 (3) Phenotype of F_1 resembles both parents
 (4) None of these
109. Which of the following is a good example of multiple allele :-
 (1) ABO blood group
 (2) Size of starch grain in pea
 (3) Shape of seed
 (4) Flower colour in pea
110. A man having the genotype AABbCcDD can produce P number of genetically different sperms, and a woman of genotype LLMMNn can generate Q number of genetically different eggs. Determine the values of P and Q.
 (1) $P = 4, Q = 4$ (2) $P = 4, Q = 8$
 (3) $P = 8, Q = 4$ (4) $P = 8, Q = 8$
111. In mice, black coat colour (allele B) is dominant to brown coat colour (allele b). The offspring of a cross between a black mouse (BB) and a brown mouse (bb) were allowed to interbreed. What percentage of the progeny would have black coats ?
 (1) 25 % (2) 50 %
 (3) 75 % (4) 100 %
112. Among the seven pairs of contrasting traits in pea plant as studied by Mendel, the number of traits related to flower and pod.
 (1) 2, 2 (2) 2, 3
 (3) 1, 2 (4) 1, 1
113. Fruit shape in shepherd's purse (Capsella bursa) is of two types—triangular and top-shaped. Triangular fruit shape (T) is dominant over top-shape (t). Following table summarizes the results of several crosses.
- | Cross | Result |
|----------------------|-------------------------------|
| Strain 1 $\times$ tt | All triangular |
| Strain 2 $\times$ tt | 1 triangular : 1 top – shaped |
| Strain 3 $\times$ tt | All top – shaped |
| Strain 4 $\times$ Tt | 3 triangular : 1 top – shaped |
- Which pair of strains possess the genotype Tt?
 (1) Strains 2 and 3 (2) Strains 2 and 4
 (3) Strains 1 and 3 (4) Strains 1 and 4

114. Types of gametes formed by the plant with genotype AABbccDD will be
(1) 4 (2) 16 (3) 8 (4) 2
115. Which Mendelian cross can produce two genotypes and two phenotypes
(1) Monohybrid cross
(2) Monohybrid test cross
(3) Incomplete dominance
(4) Co-dominance
116. Which one of the following crosses would have 1 : 1 : 1 : 1 ratio ?
(1) TtRR × ttrr (2) TTRR × ttrr
(3) TtRr × ttrr (4) TtRR × TTrr
117. Possible blood group in children from the parents with 'B' and 'O' blood group are :
(1) All B (2) All O
(3) Both B and O (4) A and B
118. When only one allele of a pair is present the condition is called ?
(1) Homozygous
(2) Hemizygous
(3) Heterozygous
(4) Incomplete dominance
119. Reciprocal cross is
(1) Intraspecific hybridization
(2) Back cross
(3) Pollen grains from one variety deposited on the stigma of the contrasting variety and vice-versa, in hybridization experiments
(4) Test cross
120. Complementary genes were demonstrated by Bateson and Punnett in
(1) *Capsella*
(2) *Lathyrus odoratus*
(3) *Antirrhinum*
(4) *Mirabilis*
121. Which of the following genotype of sweet pea plant is related with the production of purple coloured flowers ?
(1) CcPp (2) CCpp
(3) ccPP (4) Ccpp

122. Total number of genotypes and phenotypes in a dihybrid test cross is -
(1) 9 (2) 8 (3) 16 (4) 13
123. What is the frequency of pure round seeded pea plants in F₂ generation of a dihybrid cross -
(1) 4/16 (2) 6/16 (3) 1/16 (4) 2/16
124. What will be the phenotypic ratio of offsprings of F₂ generation obtained by selfing of F₁ plant having AABbCC genotype
(1) 3 : 1
(2) 1 : 1
(3) 0 : 3 : 3 : 1
(4) 27 : 9 : 9 : 9 : 3 : 3 : 3 : 1
125. Read the following statements and identify correct ones :-
(I) Gene that control seed shape in garden pea is a pleiotropic gene.
(II) Starch grain size in garden pea seeds is a codominant character.
(III) Seed shape in garden pea is regulated by SBE gene.
(1) I, II and III (2) I and III
(3) Only II (4) II and III
126. Find out the correct match from column I, II and III of following table :-

Cross type		Type of genotypes		Type of phenotypes	
(A)	Monohybrid	(I)	9	(1)	8
(B)	Dihybrid	(II)	27	(2)	4
(C)	Trihybrid	(III)	3	(3)	2

- (1) A – III – 3, B – I – 2, C – II – 1
(2) A – II – 1, B – III – 2, C – I – 3
(3) A – I – 2, B – III – 1, C – II – 3
(4) A – III – 3, B – II – 1, C – I – 2
127. Which of the following is not true for F₂ generation of dihybrid cross -
(1) 9-phenotypic categories, 4 genotypic categories
(2) 9-genotypic categories, 16 zygotic combination
(3) 62.5% parental combination, 37.5% new combination
(4) 4-types male gametes, 4 type female gamete

128. Mendelian monohybrid cross is not a presentation of

- (1) Unit factor
- (2) Dominance
- (3) Monogenic extranuclear inheritance
- (4) Segregation

129. In pea plant, genotype (Bb) shows -

- (1) Round seed with large sized starch grain
- (2) Round seed with intermediate sized starch grain
- (3) Round seed with small sized starch grain
- (4) Wrinkled seed with small sized starch

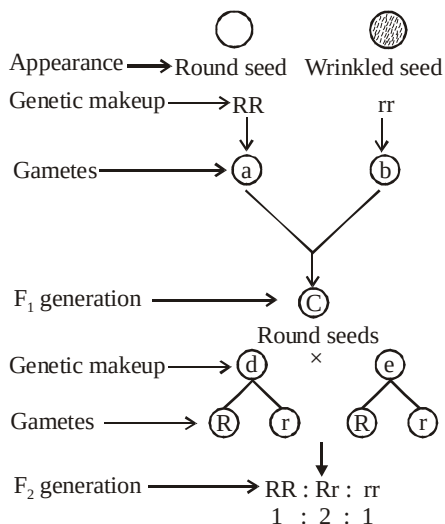
130. Which of the following is incorrect for garden pea?

	Character	Genotype	Phenotype
(1)	Flower position	Homozygous or heterozygous	Terminal
(2)	Seed colour	Homozygous or heterozygous	Yellow
(3)	Pod colour	Homozygous or heterozygous	Green
(4)	Pod shape	Only homozygous	Constricted

131. A female is heterozygous for two autosomal gene M and N and homozygous for r gene, then what will be proportion of gamete having all dominant allele :-

- (1) 1/8 (2) 1/4 (3) 1/16 (4) 0

132. Given below is the diagram of a monohybrid cross between round and wrinkled seeded pea plants. The round allele (R) is dominant over wrinkled allele (r). From the given diagram identify a, b, c, d & e :-



(1) a = R, b = r, c = RR, d = rr, e = RR

(2) a = R, b = r, c = Rr, d = rr, e = RR

(3) a = r, b = R, c = rr, d = Rr, e = RR

(4) a = R, b = r, c = Rr, d = Rr, e = Rr

133. In a certain plant red flower is dominant over white colour flower. When homozygous (RR) Plant is cross with white flower plant 200 offsprings are obtained. Then find out number of red flower offsprings.

(1) 150

(2) 200

(3) 50

(4) 100

134. Which one of the following characters chosen by Mendel can express in heterozygous condition ?

(1) Green seed colour

(2) Constricted pod shape

(3) Terminal flower position

(4) Green pod colour

135. Which statement is not correct?

(a) Male individual is hemizygous for autosomal characters.

(b) 7 Mendelian characters genes are located on chromosome number 1st, 4th, 5th and 7th.

(c) In pea plant wrinkled seeds are obtained due to deficiency of starch branching enzyme

(d) In pea plant gene of plant height & pod colour is present on chromosome-1st.

(1) b, c

(2) a, d

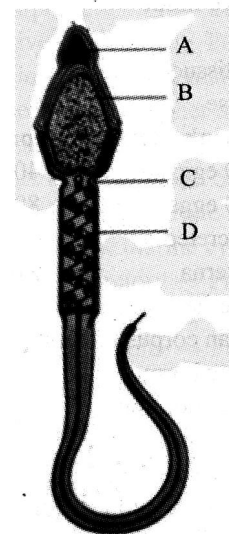
(3) a, c

(4) a, b, c, d

- | | |
|---|---|
| <p>136. Birth canal comprises of
 (1) Vaginal canal only
 (2) Cervical canal only
 (3) Fallopian tube
 (4) Vaginal canal and cervical canal</p> <p>137. The space between two folds of labia minora is called
 (1) Perineum
 (2) Vestibule
 (3) Clitoris
 (4) Foreskin</p> <p>138. In females, an erectile body homologous with penis of male is
 (1) Prepuce (2) Fourchette
 (3) Clitoris (3) Uterus</p> <p>139. Which of the following is not a male secondary sexual character?
 (1) Enlarged penis
 (2) Hoarse voice
 (3) Beard and moustache
 (4) Deposition of fats in buttocks</p> <p>140. Mammary glands are
 (1) Modified sebaceous gland
 (2) Modified sweat gland
 (3) Homologous to male prostate
 (4) Endocrinal glands</p> <p>141. Deposition of fats in the breast, hips as well as growth of pubic hair during puberty is under the influence of
 (1) Androgen
 (2) Progesterone
 (3) Estrogen
 (4) LH</p> <p>142. The vaginal opening remains partially covered by a membranous
 (1) Fimbriae
 (2) Hymen
 (3) Perineum
 (4) Vestibule</p> <p>143. Among clitoris, hymen, vagina, cervix, labia majora, labia minora and mons pubis, how many are part of external genitalia (Vulva) in females?
 (1) 3 (2) 5 (3) 2 (4) 6</p> | <p>144. Most confirmatory forensic test for rape is
 (1) Presence of glucose in vagina
 (2) Presence of hymen
 (3) Presence of fructose in vagina
 (4) Presence of LH in urine</p> <p>145. Which of the following cells undergoes second meiotic division?
 (1) Spermatogonia
 (2) Spermatids
 (3) Primary spermatocyte
 (4) Secondary spermatocyte</p> <p>146. Surgical removal of prepuce/foreskin is called
 (1) Castration
 (2) Circumcision
 (3) Cryptorchidism
 (4) Orchidectomy</p> <p>147. Energy for sperm motility is provided by
 (1) Head
 (2) Tail
 (3) Neck
 (4) Middle piece</p> <p>148. The release of sperms from seminiferous tubule is called
 (1) Semination
 (2) Spermiation
 (2) Ejaculation
 (4) Insemination</p> <p>149. Acrosomal cap of sperm is derived from
 (1) Mitochondria
 (2) Centriole
 (3) Golgi bodies
 (4) Nucleus</p> <p>150. All of the following are present in acrosome of sperm, except
 (1) Acrosin
 (2) Zonalysin
 (3) Corona penetrating enzyme
 (4) Hyaluronic acid</p> |
|---|---|

151. Which of the following undergoes the first meiotic division leading to formation of two equal, haploid cells with 23 chromosomes?
- Spermatogonia
 - Secondary spermatocyte
 - Primary spermatocyte
 - Spermatid
152. The number of sperms produced from 24 secondary spermatocytes will be
- 24
 - 12
 - 48
 - 96
153. Seminal plasma in human males is rich in
- Glucose and calcium
 - Fructose and calcium
 - Ribose and sodium
 - DNA and sodium
154. How many sperms are ejaculated by human male during a coitus?
- 200 - 300 billion
 - 20-30 million
 - 200 - 300 million
 - 500 million
155. The distal centriole gives attachment to the axial filament but the proximal helps in
- Motility
 - Cleavage of fertilized egg
 - Provides energy
 - Helps distal centriole
156. Tubuli recti connect
- Rete testis and epididymis
 - Two seminiferous tubules
 - Epididymis and vas deferens
 - Seminiferous tubule and rete testis
157. Spermiogenesis is facilitated by secretion of some factors from ____.
- Leydig cells
 - Sertoli cells
 - Interstitial cells
 - Spermatogonia
158. Seminal plasma is not contributed by
- Prostate gland
 - Seminal vesicle
 - Bulbourethral gland
 - Bartholin's gland
159. Reduced motility of sperms is called ____
- Azoospermia
 - Oligospermia
 - Asthenozoospermia
 - Teratozoospermia

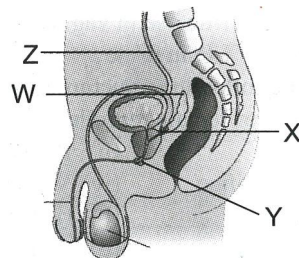
160. Which of the following is incorrect regarding semen?
- 200 - 300 million sperms are released during ejaculation.
 - Milky appearance of semen is due to secretions from prostate gland.
 - Semen is slightly acidic with pH 6.7.
 - The seminal plasma along with sperms constitute semen.
161. Choose the correct answer which shows all accurate labelling of human sperm.



- A - Acrosome, B - Neck, C - Axial filament, D - Nucleus
 - A - Acrosome, B - Proximal centriole, C - Mitochondria, D - Nucleus
 - A - Acrosome, B - Head, C - Neck, D - Mitochondria
 - A - Acrosome, B - Proximal centriole, C - Axial filament, D - Nucleus
162. The male accessory ducts include
- Rete testis, vasa efferentia, epididymis and vas deferens
 - Vasa efferentia, ejaculatory duct, urethra
 - Seminiferous tubules, vasa efferentia, epididymis, vas deferens
 - Epididymis and vasa efferentia only

163. Seminiferous tubules are lined by two types of cells
 (1) Sertoli cells and Leydig cells
 (2) Sertoli cells and Nurse cells
 (3) Spermatogonial cells and Sertoli cells
 (4) Spermatogonial cells and Leydig cells
164. The hormone responsible for the descend of testis into scrotal sac is
 (1) LH (2) Testosterone
 (3) ICSH (4) Progesterone
165. In testis, the interstitial spaces are regions outside the seminiferous tubules and contain
 (1) Leydig cells
 (2) Sertoli cells
 (3) Immunologically competent cells
 (4) Both (1) and (3)
166. The part of fallopian tube closer to ovary is
 (1) Isthmus (2) Cervix
 (3) Ampulla (4) Infundibulum
167. If vas deferens of both sides are cut, then
 (1) Gamete production is inhibited
 (2) There is no desire for sexual intercourse
 (3) Sperm count remains same in semen
 (4) Semen is without sperms
168. Which of the following is the correct path of movement of sperms?
 (1) Seminiferous tubule → vasa efferentia → rete testis → caput epididymis → tubuli recti
 (2) Seminiferous tubule → rete testis → vasa efferentia → tubuli recti → cauda epididymis
 (3) Seminiferous tubule → tubuli recti → rete testis → vasa efferentia → caput epididymis → corpus epididymis
 (4) Seminiferous tubuli → caput epididymis → vasa efferentia → corpus epididymis → rete testis

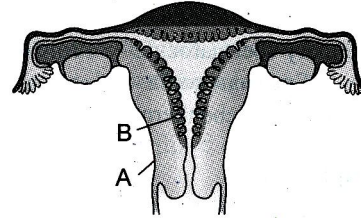
169. Mammals having intra abdominal testes are
 (1) Marsupials and whale
 (2) Cat, dog and whale
 (3) Egg laying mammals, whale and elephant
 (4) Marsupials and elephant
170. Which of the following function of Sertoli cell mediates a negative feedback control of Follicle Stimulating Hormone (FSH) secretions?
 (1) Formation of testosterone
 (2) Formation of anti-mullerian factor
 (3) Formation of blood testis barrier
 (4) Synthesis of inhibin
171. Scrotum is connected with the abdomen by
 (1) Detrusor muscles
 (2) Dartos muscles
 (3) Gubernaculum
 (4) Inguinal canal
172. The muscles present in the wall of scrotum is
 (1) Detrusor
 (2) Cremaster
 (3) Dartos
 (4) Intercostal
173. Which among the following releases secretions that helps in lubrication of the penis?



- (1) Z (2) W
 (3) X (4) Y
174. The number of mammary lobes present in each breast of human female is
 (1) 15 – 20 (2) 5 – 10
 (3) 30 – 40 (4) 50 – 70

175. In female external genitalia, a pair of large skin folds homologous with scrotum of male is -
 (1) Labia minora
 (2) Labia majora
 (3) Clitoris
 (4) Fourchette
176. In mammary glands milk is sucked out through
 (1) Ampulla
 (2) Alveoli
 (3) Lactiferous duct
 (4) Mammary ductules
177. Which of the following structure is/are present in vestibule part of female reproductive system?
 (i) Clitoris
 (ii) Perineum
 (iii) Urethral orifice
 (iv) Vaginal orifice
 (1) (i) and (ii) (2) (iii) and (iv)
 (3) (ii) and (iii) (4) (i) and (iv)
178. The cervical canal opens into vagina by an opening called
 (1) Internal Os (2) External Os
 (3) Ampulla (4) Fundus

179. Select the option which gives correct identification of labelled parts A and B along with functional characteristics in the given sectional view of female reproductive system.



- (1) A - perimetrium, composed of ciliated cuboidal epithelium
 (2) B - Myometrium, which undergoes strong uterine contraction at the time of child birth
 (3) B - Endometrium is the inner glandular lining formed of columnar epithelium
 (4) A - Myometrium, made of smooth muscles
180. Which of the following hormone controls the process of spermiogenesis?
 (1) LH (2) ICSH
 (3) FSH (3) GnRH

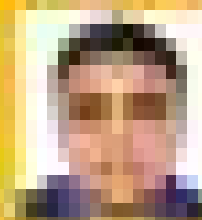
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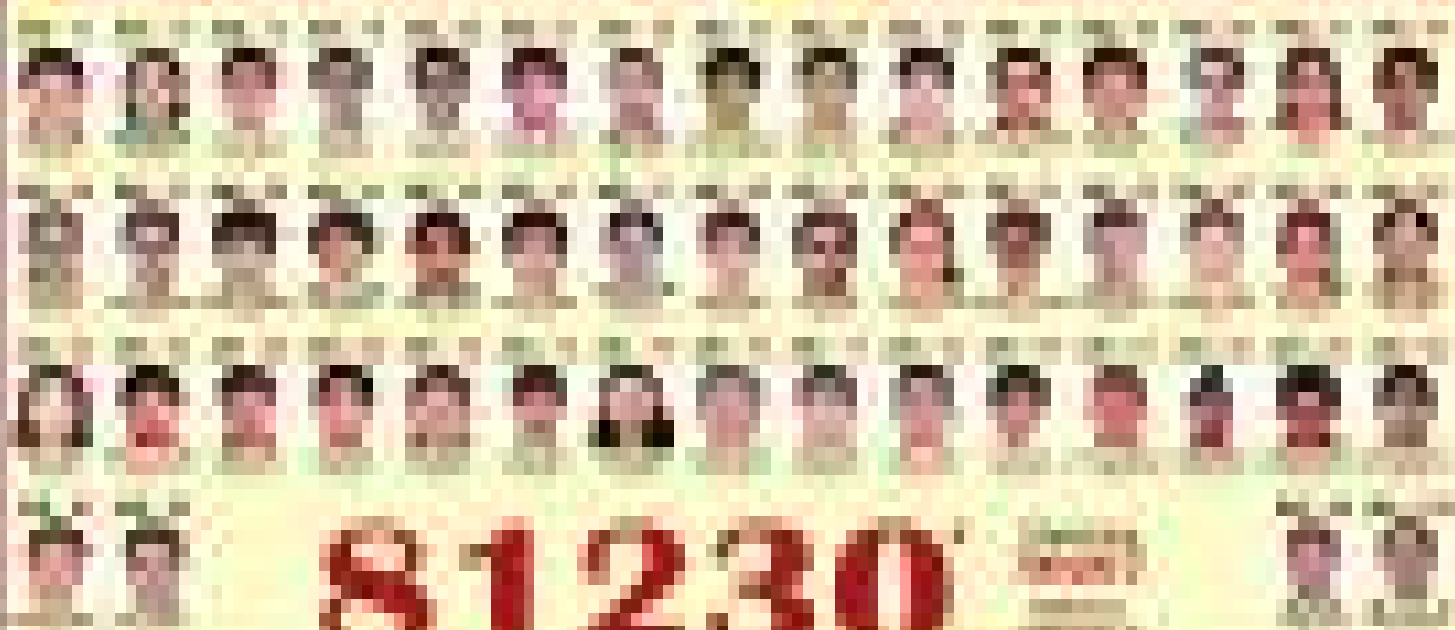


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